

SLOT SIGHT

AI Parking Predictor

Know if a spot will be free before you arrive

IBM Workout Project | Presented by: ARATHI P | S1 FT

Problem: parking wastes time and money

Up to 30%

of city traffic was estimated to be drivers searching for parking (older studies)

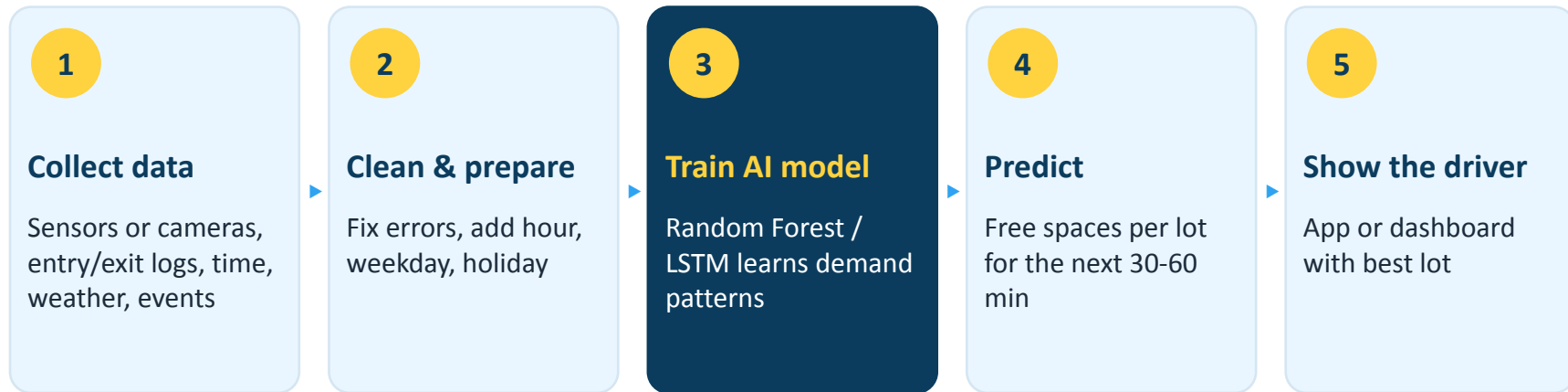
~10%

of urban traffic at peak hours is cruising for parking (US FHWA, 2023)

- Drivers circle blocks looking for a space: more traffic, fuel use and emissions.
- Most parking apps show only what is free right now. By the time you arrive, it may be full.
- Demand changes with time, weather, events and holidays.
- Result: stressed drivers, late arrivals and lost customers for shops and malls.

The gap: drivers need to know what WILL be free when they arrive. That is a prediction problem, ideal for AI.

Solution: AI predicts free spaces early

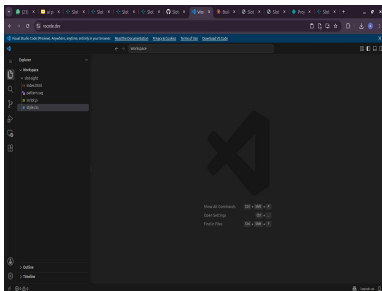


↻ *Feedback loop: real occupancy retrains the model*

How it learns: from past data it finds patterns, e.g. Monday 9 AM + rain + match day means Lot A fills by 9:30.

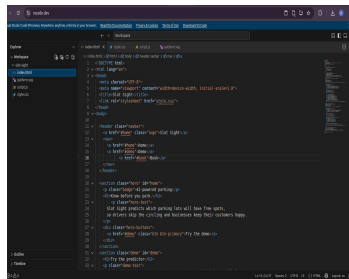
How we built the website

1 Set up



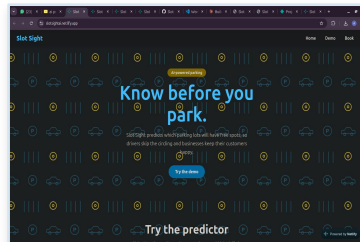
Free browser editor
(vscode.dev); created the
slot-sight folder

2 Structure



index.html: navbar, hero,
demo, booking form,
footer

3 Style



style.css: sky blue +
sunny yellow, rounded
cards

How the project works

1

Pick the situation

Choose time, weather and special event (rush hour, sale day, match day, concert...)

2

Press Predict

The site applies sample rules to three lots: A Main Entrance, B Basement, C Rooftop

3

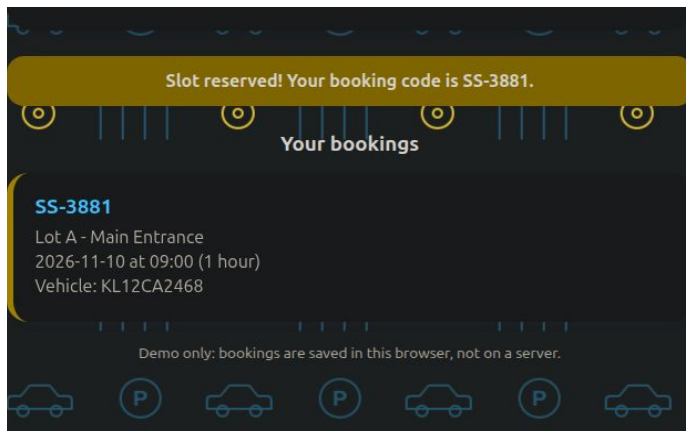
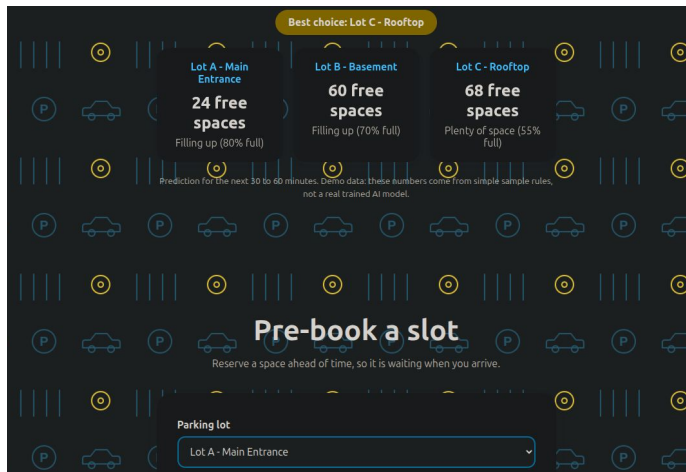
See the result

Each lot shows its status; the best lot gets a yellow "Best choice" tag

4

Pre-book a slot

Reserve a lot, date and time; get a booking code like SS-1234



Results

3

sample lots compared in one click

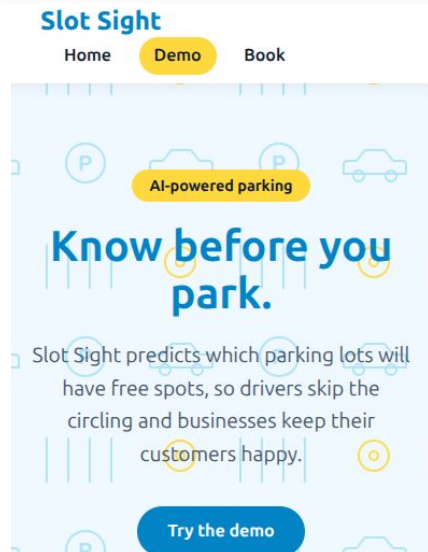
2

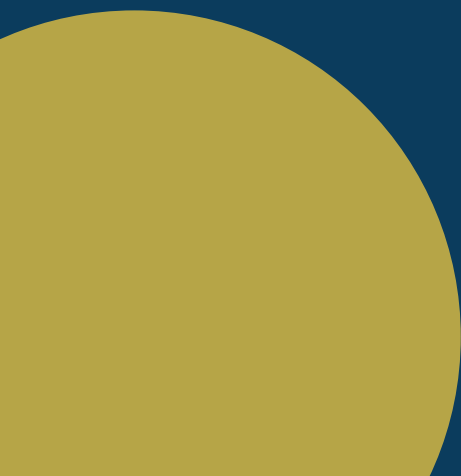
working features: predict + pre-book

Free

tools only, built from zero

- A working website that shows a parking prediction and a best-lot suggestion
- Pre-booking with a booking code, saved in the browser (demo)
- Checked on a phone screen; made for malls, offices and event venues
- Next: real parking data, a trained AI model, and putting the site online free





Thank you

Slot Sight: know before you park

Questions?